

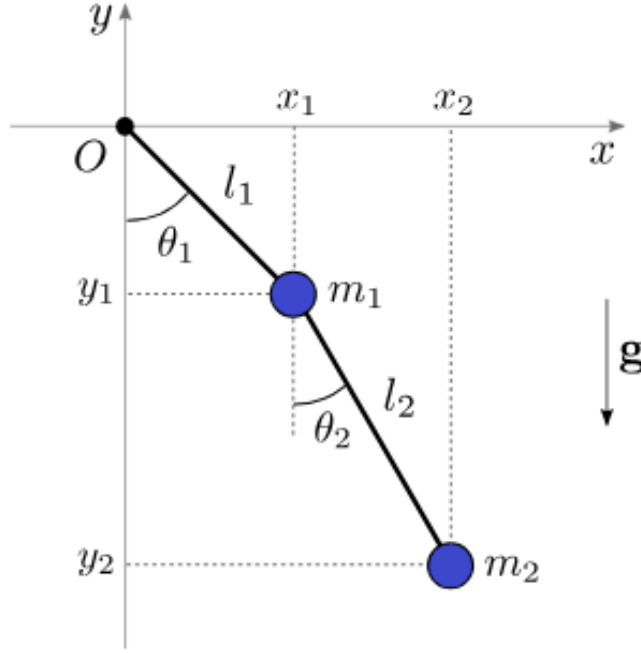


Figura 1: Diagram (from Diego Assencio)

Given that the double pendulum is a chaotic system, a first-order method such as Euler's method can be sufficient to represent its qualitative erratic motion. Although higher-order numerical methods provide greater accuracy over short time intervals, in a chaotic system small numerical errors grow exponentially over time. Therefore, after a sufficiently long time, even highly accurate numerical methods will eventually diverge from the exact trajectory of the real system.

In the following deduction, we will find the formulas we need to implement a python simulation to represent a double pendulum using this method.

We will assume the simplification that $l_1 = l_2 \equiv l$ and $m_1 = m_2 \equiv m$. Also, we will be using the following angles to simplify the reading between this paper's results and the Python code, $\alpha = \theta_1$ and $\beta = \theta_2$.

First of all, we must express our cartesian coordinates in terms of our two generalized ones.

$$\begin{aligned} x_1 &= l \sin \alpha & x_2 &= l (\sin \alpha + \sin \beta) \\ y_1 &= -l \cos \alpha & y_2 &= -l (\cos \alpha + \cos \beta) \end{aligned}$$

We can also get the velocities, which are just the coordinates time derivatives.

$$\begin{aligned} \dot{x}_1 &= \dot{\alpha} l \cos \alpha & \dot{x}_2 &= l (\dot{\alpha} \cos \alpha + \dot{\beta} \cos \beta) \\ \dot{y}_1 &= \dot{\alpha} l \sin \alpha & \dot{y}_2 &= l (\dot{\alpha} \sin \alpha + \dot{\beta} \sin \beta) \end{aligned}$$

Their square value will also be useful to easily replace in the kinetic energy formula.

$$\begin{aligned} \dot{x}_1^2 &= \dot{\alpha}^2 l^2 \cos^2 \alpha & \dot{x}_2^2 &= l^2 (\dot{\alpha}^2 \cos^2 \alpha + \dot{\beta}^2 \cos^2 \beta + 2\dot{\alpha}\dot{\beta} \cos \alpha \cos \beta) \\ \dot{y}_1^2 &= \dot{\alpha}^2 l^2 \sin^2 \alpha & \dot{y}_2^2 &= l^2 (\dot{\alpha}^2 \sin^2 \alpha + \dot{\beta}^2 \sin^2 \beta + 2\dot{\alpha}\dot{\beta} \sin \alpha \sin \beta) \end{aligned}$$

We can define the Lagrangian of our system by using the well known formulas for the kinetic and gravitational potential energies, .

$$\begin{aligned} K &= \frac{1}{2}m(\dot{x}_1^2 + \dot{y}_1^2 + \dot{x}_2^2 + \dot{y}_2^2) = ml\dot{\alpha}^2 + \frac{1}{2}ml\dot{\beta}^2 + ml\dot{\alpha}\dot{\beta} \cos(\alpha - \beta) \\ U &= mg(y_1 + y_2) = -2mgl \cos \alpha - mgl \cos \beta \\ \Rightarrow L \equiv K - U &= ml\dot{\alpha}^2 + \frac{1}{2}ml\dot{\beta}^2 + ml\dot{\alpha}\dot{\beta} \cos(\alpha - \beta) + 2mgl \cos \alpha + mgl \cos \beta \end{aligned}$$

The equations of movement are given by Lagrange's Equation's solutions. In this system both lagrange equations are coupled and are $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\alpha}} \right) - \frac{\partial L}{\partial \alpha} = 0$ and $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\beta}} \right) - \frac{\partial L}{\partial \beta} = 0$. We can first substitute all the terms in the first equation and simplify it.

$$\begin{cases} \frac{\partial L}{\partial \dot{\alpha}} = 2ml\dot{\alpha} + ml\dot{\beta} \cos(\alpha - \beta) \\ \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\alpha}} \right) = 2ml\ddot{\alpha} + ml\ddot{\beta} \cos(\alpha - \beta) - ml\dot{\beta} \sin(\alpha - \beta)(\dot{\alpha} - \dot{\beta}) \\ \frac{\partial L}{\partial \alpha} = -ml\dot{\alpha}\dot{\beta} \sin(\alpha - \beta) - 2mgl \sin \alpha \end{cases} \Rightarrow \boxed{2\ddot{\alpha} + \ddot{\beta} \cos(\alpha - \beta) + \dot{\beta} \sin(\alpha - \beta) + 2\frac{g}{l} \sin \alpha = 0}$$

By doing the same with the second equation,

$$\begin{cases} \frac{\partial L}{\partial \dot{\beta}} = m\dot{\beta} + m\dot{\alpha} \cos(\alpha - \beta) \\ \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\beta}} \right) = m\ddot{\beta} + m\ddot{\alpha} \cos(\alpha - \beta) - m\dot{\alpha} \sin(\alpha - \beta)(\dot{\alpha} - \dot{\beta}) \\ \frac{\partial L}{\partial \beta} = -m\dot{\alpha} \dot{\beta} \sin(\alpha - \beta) - mgl \sin \beta \end{cases} \Rightarrow \boxed{\ddot{\beta} + \ddot{\alpha} \cos(\alpha - \beta) - \dot{\alpha} \sin(\alpha - \beta) + \frac{g}{l} \sin \beta = 0}$$

To apply Euler's numeric method we need to find $\frac{d\dot{\alpha}}{dt}$ and $\frac{d\dot{\beta}}{dt}$, which are just $\ddot{\alpha}$ and $\ddot{\beta}$ respectively. We can get $\ddot{\alpha}$ with dependencies just in angles and angular velocities by substituting $\ddot{\beta}$ from the second equation into the first one.

$$\begin{aligned} \ddot{\beta} &= -\ddot{\alpha} \cos(\alpha - \beta) + \dot{\alpha} \sin(\alpha - \beta) - \frac{g}{l} \sin \beta \\ \Rightarrow 2\ddot{\alpha} + \left(-\ddot{\alpha} \cos(\alpha - \beta) + \dot{\alpha} \sin(\alpha - \beta) - \frac{g}{l} \sin \beta \right) \cos(\alpha - \beta) + \dot{\beta} \sin(\alpha - \beta) + 2\frac{g}{l} \sin \alpha &= 0 \\ \rightarrow \ddot{\alpha}(2 - \cos(\alpha - \beta)) + \sin(\alpha - \beta)(\dot{\alpha} \cos(\alpha - \beta) + \dot{\beta}) + \frac{g}{l}(2 \sin \alpha - \cos(\alpha - \beta) \sin \beta) & \end{aligned}$$

By isolating $\ddot{\alpha}$ and using the numeric result to input it into $\ddot{\beta}$, we already have both functions we need to apply Euler's method.

$$\boxed{\frac{d\dot{\alpha}}{dt} = \left[\sin(\alpha - \beta)(\dot{\alpha} \cos(\alpha - \beta) + \dot{\beta}) + \frac{g}{l}(2 \sin \alpha - \cos(\alpha - \beta) \sin \beta) \right] \frac{1}{\cos(\alpha - \beta) - 2}}$$

$$\boxed{\frac{d\dot{\beta}}{dt} = -\ddot{\alpha} \cos(\alpha - \beta) + \dot{\alpha} \sin(\alpha - \beta) - \frac{g}{l} \sin \beta}$$